

Does the ranking of your school matter for a biosciences PhD?

Testing the question against the pharma R&D labour market, with Pfizer as the worked case — what the published evidence settles, what it does not, and a reproducible way to measure the rest.

54.1%

of US life-science PhDs
head to industry (2022)

~14%

hold a tenure-track post
5-6 years after the PhD

459

US institutions award
research doctorates

~70%

of US biotech jobs sit in
Boston, San Diego, RTP

The short answer

Yes — you can be hired as a scientist at Pfizer with a PhD from a non-elite school, and the structure of the market says this is ordinary rather than exceptional.

Three things make the biosciences case different from the prestige-gated markets people have in mind. First, the modal destination for a US life-science PhD is now industry, not academia — 54.1% of 2022 graduates reported heading into industry against 26.9% into academia, an almost exact inversion of 1992. Industry is the main market, not the fallback, and it is far larger than the number of faculty posts that the prestige hierarchy rations.

Second, the thing pharma R&D hires for is a technique and a therapeutic area, not a transcript. A posting for a cryo-EM structural biologist, an ADC chemist or a CAR-T immunologist is filled by whoever has run that bench, and that population is small enough in any given modality that filtering it further by school name would leave too few candidates to fill the role.

Third, and most important: a bioscience PhD is followed by a postdoc, and the postdoc re-rates you. Publication record and postdoc lab are earned after the PhD admission decision, they are what a hiring scientist actually reads, and they are the reason the PhD institution is a weaker signal here than in fields with no such second stage.

What this does *not* say is that school never matters. It matters at the margins that the report sets out in section 5 — access to the postdoc, geography, and the recruiting network — and the honest position on the central quantity is that no published study measures institution-level intake into pharma R&D. Section 6 is a pipeline that measures it.

1. Why this differs from the economics case you raised

The comparison is a good one, because economics is close to the opposite configuration. An economics PhD has no postdoc stage: you go on the job market straight out of the programme, so the department's name and your advisor's letters are doing most of the signalling work, and they are doing it on the single occasion that sets your starting position. Placement records show the effect — at MIT, 32% of academic placements went to top-15 US economics departments or top-10 business schools, a rate no mid-ranked department approaches.

And the pattern holds on the private-sector side too: the tech, central bank and consulting employers who hire economics PhDs are reading the same department-plus-letters signal, because there is no wet-lab skill portfolio to read instead.

The irony is that the economics literature contains the sharpest available evidence *against* taking department rank as a proxy for individual quality. Conley and Önder tracked 14,299 economics PhDs from 154 institutions and found that the median graduate of a top-ten department had fewer than 0.03 AER-equivalent publications six years out — an untenurable record almost anywhere — while Carnegie Mellon's 85th-percentile graduates out-published the 85th percentile of Chicago, Penn, Stanford and Berkeley. Rank within your cohort predicts research output; the rank of the department predicts it surprisingly poorly.

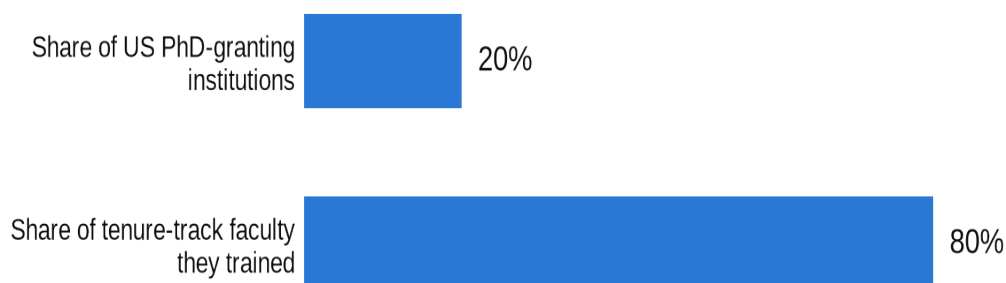
So your instinct that economics has a broad band of respectable programmes is right on the merits and wrong about the market: the programmes are substitutable in what they teach, but the academic market prices them as if they were not. Biosciences industry is the case where the market has a better instrument — your actual published work.

2. The benchmark: what a genuinely prestige-gated market looks like

Before asking whether pharma screens on pedigree, it helps to have a calibrated example. Academic faculty hiring is one. Wapman, Zhang, Clauzet and Larremore assembled the near-complete US tenure-track faculty census for 2011–2020 and found that roughly a fifth of PhD-granting institutions supplied about four fifths of all tenure-track faculty, with five universities alone training one in eight of the faculty at doctoral institutions. Clauzet's earlier work put 71–86% of faculty as coming from the top quarter of institutions.

The academic market, for scale

A fifth of the schools trained four fifths of US tenure-track faculty



Source: Wapman et al., *Nature* 610 (2022), 120–127. Shares are approximate, across all fields.

Lauren Rivera's interview study of elite investment banks, law firms and consultancies is the other calibrated example, and a starker one: there, school prestige was the single most important résumé screen, recruiting was restricted to a handful of 'super-elite' campuses, and interviewers attributed superior

ability to candidates from those schools irrespective of their actual performance.

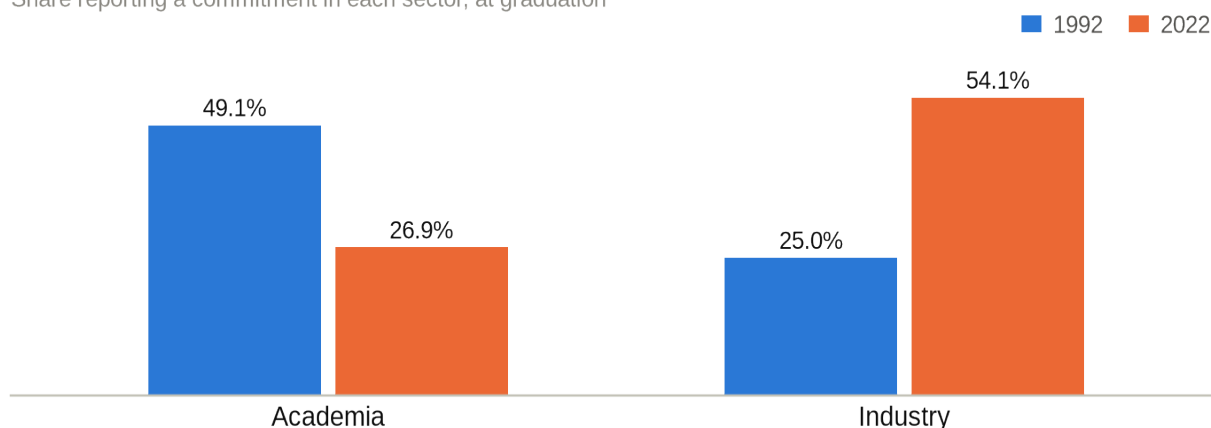
These are the shapes to compare against. Neither resembles the stated requirements of a pharma bench-science posting.

3. What the evidence says about the biosciences case

3.1 Industry is now the majority destination

Where US life-science PhDs said they were heading

Share reporting a commitment in each sector, at graduation



Source: NSF NCSSES Survey of Earned Doctorates, post-graduation commitments of US life-science doctorate recipients, 1992 and 2022.

The academic route that the prestige hierarchy rations has been shrinking as a share of outcomes for thirty years. Roughly 14% of biological sciences PhDs hold a tenure-track post 5–6 years after graduating and about 24% by ten years out, against the 78% of graduate students who report wanting an academic research career. Whatever prestige does to faculty hiring, it is operating on a progressively smaller slice of where bioscience PhDs actually go.

3.2 Where prestige effects have been measured, they attach to the academic path

The quantitative work that exists separates the two markets rather cleanly. A study of engineering doctorates using 2015–2018 Survey of Earned Doctorates data found the research intensity of the PhD institution significantly predicted choosing an *academic* career path. A study of philosophy PhDs found that once career *preference* was controlled for, institutional prestige was no longer significantly associated with academic-versus-non-academic employment — that is, much of the apparent prestige effect was graduates of elite programmes wanting academic jobs more, not being uniquely able to get non-academic ones.

Neither is a biosciences industry study, and that is the honest state of the literature: the sign of the effect for industry hiring is not established by direct measurement.

3.3 What the employer actually asks for

Pfizer's own Senior Scientist postings state the requirement as a PhD or equivalent in a named discipline — 'biochemistry, structural biology, or a related field' — with no institutional condition, at advertised bands around \$88–152k across the Groton, La Jolla and Cambridge sites. Its internal postdoctoral programme runs 40-plus positions across roughly a dozen campuses, which is a second, publication-based entry route into the company that is open after the PhD is finished.

Stated requirements are weak evidence about revealed preference — no employer advertises a pedigree filter it applies informally. They do establish that the filter is not formal, which matters for whether an application is worth making.

4. The mechanism: a technique market, not a pedigree market

- **Modality-specific demand.** Requisitions are written around a platform — protein degraders, mRNA formulation, gene editing delivery, DEL chemistry. The qualified population per modality is small, so screening it further on school name is a luxury the requisition cannot afford.
- **The postdoc re-rates you.** Publication record and postdoc lab are acquired after the PhD, are legible to the hiring scientist, and dominate the read. This is the structural feature economics lacks.
- **The hiring manager is a scientist.** Pharma R&D panels are staffed by bench scientists reading your papers and quizzing your methods, not by HR screening on institution tier — the opposite of the campus-recruiting model Rivera documented in finance and law.
- **Geography beats ranking.** Around 70% of US biotech employment sits in Boston, San Diego and the Research Triangle, and candidates rarely move between clusters without a relocation conversation. Being at a mid-ranked school inside a cluster can beat an elite school outside one.
- **Salary bands compress the return.** At an advertised \$88–152k for Senior Scientist, there is no pedigree premium of the kind that makes elite-school screening profitable for a bank.

5. Where school does still bite

The defensible version of the prestige claim is that it operates indirectly, and mostly before the industry application:

- **Postdoc access.** If the competitive postdoc is the re-rating mechanism, then PhD institution matters to the extent it gates entry to a strong postdoc lab. This is the main channel and it is real.
- **Advisor network.** Referrals do a lot of hiring. A well-connected advisor with industry collaborators is worth more than the institution name, and such advisors are not evenly distributed — but they are much less concentrated than institutional rank.
- **Cluster proximity.** Internships, collaborations and part-time industry contact accrue to students near a hub.
- **Leadership tiers.** The senior R&D and CSO population looks more elite-trained than the bench population. Whether that is hiring, promotion or survivorship is exactly what section 6's seniority gradient is designed to separate — and they are different claims about different decisions.
- **Resource confound.** Better-funded programmes offer more expensive techniques and more publication support. The effect is real but it is about the training, not the name, and it varies far more by lab than by institution.

6. Measuring the part nobody has published

The gap in the evidence is an institution-level estimate of who actually gets hired into pharma R&D. It is measurable from open data, and `pedigree_pipeline.py` in this directory implements it. The design point that matters most is the denominator.

Elite schools award a large share of all doctorates. So ‘most of our scientists went to a top-30 school’ is uninterpretable on its own — it has to be divided by the share of PhDs those schools produce.

The estimand is therefore a representation ratio per institution tier: the employer's share of scientists trained in that tier, divided by that tier's share of US bio/biomedical doctorates awarded. A ratio of 1 means the intake looks exactly like the applicant pool — no prestige premium. Above 1 means over-representation, and the size of the excess is the answer to your question.

Step	What it does	Source
1. Resolve employer	Union every OpenAlex institution record for the employer, so site-level and subsidiary records are not dropped.	OpenAlex /institutions
2. Pull scientists	Cursor-page all authors whose last known affiliation is the employer; keep those whose topics sit in Life or Health Sciences.	OpenAlex /authors
3. Infer training site	Take the earliest dated education-type affiliation in the author's history. Flag records where a second academic affiliation starts within 4 years (PhD vs postdoc not separable).	OpenAlex affiliations
4. Tier institutions	Rank US doctorate-granting institutions on life-science output, and again on NIH award dollars. A result must survive both axes.	OpenAlex /works, NIH RePORTER
5. Get the denominator	Bio/biomedical doctorates awarded per institution - the applicant pool the employer draws from.	NSF NCSES SED table 7-3
6. Representation ratio	Employer share divided by PhD-production share, per tier, with a percentile bootstrap over authors.	computed
7. Validate the rule	Hand-label a random sample of 100 inferred PhD institutions against public CVs and report the error rate.	manual

Training institution is reconstructed from OpenAlex author records, which carry a dated affiliation history, so the earliest education-type affiliation stands in for the PhD institution. OpenAlex is CC0 and has a public API. LinkedIn is deliberately not used: scraping it breaches its terms and its self-reported education fields are unverifiable.

Comparators matter as much as Pfizer itself. Running the same pipeline over Merck, AstraZeneca, Genentech, Amgen, Regeneron, Vertex and Moderna turns a single company's number into a distribution, and it is the spread across employers — not any one value — that tells you whether pedigree screening is an industry norm or a house style.

What would make the result wrong

The pipeline reports these rather than burying them, and the report's conclusion should be read as conditional on them:

- **Publication bias in the sample.** Only scientists who publish under a Pfizer affiliation are visible. Process development, regulatory and much of clinical operations are not, and they are a large share of R&D headcount — plausibly the share where pedigree matters least, which would bias the measured premium upward.

- **Thin affiliation history before ~2010.** Senior staff are under-sampled relative to junior staff.
- **PhD versus postdoc.** The two are not reliably separable from publication metadata; the run is reported with and without the ambiguous records.
- **Undergraduate co-authorship.** A paper published before the PhD would misattribute the training institution.
- **Survivorship.** This measures who is employed now, not who was hired, so it cannot by itself separate hiring from retention.
- **Ranking choice.** Any tiering imports its own bias, which is why two independent axes are run and only findings surviving both are reported.

A note on why this report has no Pfizer numbers in it

The pipeline was written and unit-tested here but could not be run: the environment this report was produced in permits outbound network access to package registries only, so `api.openalex.org` is unreachable. Running the four commands in the README from a machine with ordinary internet access produces the tier table, the representation ratios and the seniority gradient. Reporting invented figures in the meantime would have been worse than reporting none.

7. Practical read

- A non-elite bioscience PhD is not a barrier to a Pfizer scientist role. The formal requirement is the degree in a relevant field.
- Optimise for the advisor, the technique and the publication record over the institution's ranking — that is what is read, and Conley and Önder's result on how weakly department rank predicts individual output generalises.
- Treat the postdoc as the real positioning decision. It is the stage where an unfashionable PhD can be re-rated, and it comes after the admissions decision you may already regret.
- Weight location heavily. A mid-ranked programme inside the Boston, San Diego or RTP cluster has an access advantage over an elite programme outside one.
- Be more cautious about the academic path. That is the market where the prestige hierarchy is documented, steep, and unlikely to move.

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